

Continuous-Time Linear Time-Invariant (LTI) System

沈威宇

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1 Continuous-Time Linear Time-Invariant (LTI) System

I Definition

i Causal continuous-time linear time-invariant (LTI) system

A causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation, given initial condition $y(0) = 0$. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$ and is such that $h(t) = 0$ for all $t < 0$, that

$$y(t) = (x * h)(t),$$

where

$$(x * h)(t) := \int_0^t x(\tau)h(t - \tau) d\tau.$$

Applying unilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function (轉移函數).

ii Non-causal continuous-time linear time-invariant (LTI) system

A non-causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$, that

$$y(t) = (x * h)(t) := \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

Applying bilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function.

iii Gain

$|H(s)|$ is called gain (增益), or attenuation (衰減量) if < 1 .

iv Order

The number of poles of $H(s)$ is called the order of the system.

v Poles and zeros

Since $H(s)$ is rational, a complex conjugate of a pole is a pole of same order, and a complex conjugate of a zero is a zero of same order.

A pole-zero plot or pole-zero constellation is a graphical representation showing poles and zeros of a transfer function in the complex plane.

vi Transient response

A response component that converges to an equilibrium as t gets large. Such converged state is called steady state or transient state.

vii Time constant

For a response function in the form $y(t) = y_0 e^{-\alpha t}$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\frac{1}{\alpha}$, that is, $y(\tau) = y_0 e^{-1}$.

For a response function in the form $y(t) = y_0(1 - e^{-\alpha t})$ with constants $y_0 \neq 0$ and $\alpha > 0$, the time constant τ is defined as $\frac{1}{\alpha}$, that is, $y(\tau) = y_0(1 - e^{-1})$.

viii Natural frequency

For a pair of conjugate poles $\pm i\omega$ of $H(s)$, ω is called a natural frequency of the system.

ix Resonance

Resonance occurs when the input and transfer function have common poles on the imaginary axis. Such natural frequency is called resonant/resonance frequency.

II Stability

i Asymptotically stable or bounded-input bounded-output (BIBO) stable

A LTI system is called asymptotically stable or bounded-input bounded-output stable iff bounded input implies bounded response, iff all poles of it are on the left half-plane iff $h(t)$ is absolutely integrable.

ii (Lyapunov) stable

A LTI system is called (Lyapunov) stable iff there is no pole of it on the right half-plane and no multiple pole of it on the imaginary axis iff $h(t)$ is bounded.

iii Marginally stable

A LTI system is called marginally stable iff it is stable but not asymptotically stable.

iv Unstable

A LTI system is called unstable iff it is not stable.

III Frequency response

i Frequency response

Assume the system is asymptotically stable. When the input is $x(t) = e^{i\omega t}$, $X(s) = \frac{1}{s - i\omega}$, the response is

$$Y(s) = \frac{H(s)}{s - i\omega}.$$
$$y(t) = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\gamma - iT}^{\gamma + iT} \frac{H(s)e^{st}}{s - i\omega} ds, \quad \gamma > 0.$$

Since $H(s)$ is meromorphic, by residue theorem,

$$y(t) = \text{Res}(Y, i\omega) + \sum_{k=1}^n \text{Res}(Y, a_k)$$

$$\text{Res}(Y, i\omega) = \lim_{s \rightarrow i\omega} (s - i\omega) \frac{H(s)e^{st}}{s - i\omega} = H(i\omega)e^{i\omega t}$$

The $\sum_{k=1}^n \text{Res}(Y, a_k)$ is the transient response. This does not work for marginally stable systems since $\sum_{k=1}^n \text{Res}(Y, a_k)$ term is not transient for them.

$$H(i\omega)e^{i\omega t}$$

is called steady-state (sinusoidal) response.

- $H(i\omega)$ is called frequency response,
- $a(\omega) := |H(i\omega)|$ is called magnitude/amplitude response/ratio or gain,
- $\theta(\omega) := \arg(H(i\omega))$ is called phase (angle) response/shift.

ω is called angular frequency and $f := \frac{\omega}{2\pi}$ is called non-angular frequency. Both can be called frequency and can be differentiated with symbol, and angular frequency is used below.

ii Normalizing

For a transfer function $H(s)$ such that

$$\max_{\omega} |H(i\omega)| = K \in \mathbb{R}_{>0},$$

normalizing means dividing $H(s)$ by K . The following filter definitions are normalized. Non-normalized definition is defined as any filter whose normalized form complies with the definition.

iii Logarithmic scale, decibel (dB), and Bode plot (波德圖)

Frequency response are often expressed in logarithmic scale:

- Magnitude response, input, and output can be expressed in decibel (dB) as $g(\omega) := 20 \log\left(\frac{|H(i\omega)|}{\max_{\omega} |H(i\omega)|}\right)$, $20 \log(x)$, and $20 \log(y)$.

- Changing by 2 times is called an octave or oct, and changing by 10 times is called a decade or dec. $k \text{ dec}^{-1}$ equals to $\log_{10}(2) \text{ oct}^{-1}$. The most common used is 20 dB/dec equals to $20 \log_{10}(2) \approx 6 \text{ dB/oct}$.
- Bode magnitude plot or dB gain curve: Magnitude response plotted in logarithmic (frequency)-logarithmic (magnitude response) scale.
- Bode phase plot or phase curve: Phase response is often plotted in logarithmic (frequency)-linear (phase response) scale.
- Asymptotic Bode plot: The asymptotes of Bode plots for low and high frequency is often interested.

We can factor $H(s)$ into

$$H(s) = G \prod_{k=1}^m H_k(s)$$

then

$$|H(i\omega)| = |G| \prod_{k=1}^m |H_k(i\omega)|$$

$$\log_{10} |H(i\omega)| = \log_{10} |G| + \sum_{k=1}^m \log_{10} |H_k(i\omega)|$$

$$\arg(H(i\omega)) = \arg(G) + \sum_{k=1}^m \arg(H_k(i\omega))$$

iv Cutoff (截止), break, or corner (angular) frequency

For a transfer function, cutoff frequency ω_c is a frequency such that

$$|H(i\omega_c)| = \frac{\max_{\omega} |H(i\omega)|}{\sqrt{2}}.$$

$20 \log_{10} \left(\frac{1}{\sqrt{2}} \right) = -10 \log_{10}(2) \approx -3$, thus this frequency is also denoted $\omega_{-3\text{dB}}$ or $\omega_{3\text{dB}}$.

v Low-pass or lowpass (低通) (LP), high-cut, or bass-cut filter (LPF)

A continuous-time linear time-invariant system is a normalized low-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\lim_{\omega \rightarrow 0} |H(i\omega)| = 1$$

$$\lim_{\omega \rightarrow \infty} |H(i\omega)| = 0$$

K is called low-pass gain.

Ideal normalized low-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega < \omega_0 \\ 0, & \omega > \omega_0 \end{cases}$$

vi High-pass or highpass (高通) (HP), low-cut, or treble-cut filter (HPF)

A continuous-time linear time-invariant system is a normalized high-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\lim_{\omega \rightarrow 0} |H(i\omega)| = 0$$

$$\lim_{\omega \rightarrow \infty} |H(i\omega)| = 1$$

K is called high-pass gain.

Ideal normalized high-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega > \omega_0 \\ 0, & \omega < \omega_0 \end{cases}$$

vii Band-pass or banetwork.captive-portal-service.enabledndpass (BP) filter (BPF)

A continuous-time linear time-invariant system is a standard normalized band-pass filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\lim_{\omega \rightarrow 0} |H(i\omega)| = 0$$

$$\lim_{\omega \rightarrow \infty} |H(i\omega)| = 0$$

$$\arg_{\omega} \left\{ |H(i\omega)| = \frac{1}{\sqrt{2}} \right\} = \omega_L, \omega_H$$

$$|H(\sqrt{\omega_L \omega_H})| = 1$$

where $\omega_L < \omega_H$ are called band edge, $\omega_H - \omega_L$ is called bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central passed frequency or center frequency. $\omega_0 \gg B$ is called narrow band-pass filter.

K is called band-pass or midband gain.

Ideal normalized band-pass filter is

$$H(i\omega) = \begin{cases} 1, & \omega_H > \omega > \omega_L \\ 0, & \omega < \omega_L \vee \omega > \omega_H \end{cases}$$

viii Band-stop or bandstop (BS), band-reject, or notch filter (BSF)

A continuous-time linear time-invariant system is a standard normalized band-stop filter iff $H(i\omega)$ as a function of a real variable ω is such that

$$\arg_{\omega} \left\{ |H(i\omega)| = \frac{1}{\sqrt{2}} \right\} = \omega_L, \omega_H$$

$$|H(\sqrt{\omega_L \omega_H})| = 0$$

where $\omega_L < \omega_H$ are called band edge, $\omega_H - \omega_L$ is called bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central rejected frequency or center frequency. $\omega_0 \gg B$ is called narrow band-stop filter.

K is called notch gain.

Ideal normalized band-stop filter is

$$H(i\omega) = \begin{cases} 0, & \omega_H > \omega > \omega_L \\ 1, & \omega < \omega_L \vee \omega > \omega_H \end{cases}$$

ix Normalized low-pass prototype filter

A normalized low-pass prototype filter is a low-pass filter with cutoff frequency $\omega_c = 1$. Suppose the transfer function of it is $H_P(s)$. A normalized low-pass prototype filter is usually defined with $|H(i\omega)|^2 = H(i\omega)H(-i\omega)$ or $H(s)$.

To transform an n th-order normalized low-pass prototype filter to an n th-order low-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{LP}(s) = H_P\left(\frac{s}{\omega_c}\right).$$

To transform an n th-order normalized low-pass prototype filter to an n th-order high-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{HP}(s) = H_P\left(\frac{\omega_c}{s}\right).$$

To transform an n th-order normalized low-pass prototype filter to a $2n$ th-order standard band-pass filter with cutoff frequency ω_c , use the transfer function

$$H_{BP}(s) = H_P\left(\frac{s^2 + \omega_0^2}{Bs}\right).$$

To transform an n th-order normalized low-pass prototype filter to a $2n$ th-order standard band-stop filter with cutoff frequency ω_c , use the transfer function

$$H_{BS}(s) = H_P\left(\frac{Bs}{s^2 + \omega_0^2}\right).$$

x Butterworth filter or maximally flat magnitude filter

n th-order Butterworth low-pass prototype filter is

$$|H(i\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

$$H(s) = \prod_{k=1}^n \frac{1}{s - e^{\frac{j(2k+n-1)\pi}{2n}}}$$

When not specified, a filter of a type usually refers to the lowest-order Butterworth filter of the type.

IV Transfer function factors

Variables after ; are parameters, which are positive real numbers unless otherwise specified.

i Ramp factor

$$H_r(s; W) := \frac{s}{W}$$

$$H_r(i\omega; W) = \frac{i\omega}{W}$$

$$|H_r(i\omega; W)| = \frac{\omega}{W}$$

$$g_r(\omega; W) = 20 \log_{10}(\omega) - 20 \log_{10}(W)$$

Bode magnitude plot has slope 20 dB/dec, y intercept $-20 \log_{10}(W)$, and x intercept $\log_{10}(W)$.

The cutoff frequency is:

$$|H_r(i\omega_c; W)| = \frac{1}{\sqrt{2}}$$

$$\omega_c = \frac{W}{\sqrt{2}}$$

$$\theta_r(\omega; W) = \frac{\pi}{2}$$

ii Low-pass factor

$$H_{LP}(s; W) := \frac{W}{s + W}$$

$$H_{LP}(i\omega; W) = \frac{W}{i\omega + W}$$

$$|H_{LP}(i\omega; W)| = \frac{W}{\sqrt{\omega^2 + W^2}}$$

$$g_{LP}(\omega; W) = 20 \log_{10}(W) - 10 \log_{10}(\omega^2 + W^2)$$

$$\theta_{LP}(\omega; W) = -\arg(i\omega + W)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \gg W$:

$$g_{LP}(\omega; W) \approx 20 \log_{10}(W) - 20 \log_{10}(\omega),$$

which is slant asymptote with slope -20 dB/dec, y intercept $20 \log_{10}(W)$, and x intercept $\log_{10}(W)$.

$$\theta_{LP}(\omega; W) \approx 0.$$

- For $\omega = W$:

$$g_{LP}(W; W) = 20 \log_{10}(W) - 10 \log_{10}(2W^2) = -10 \log_{10}(2).$$

Thus the cutoff frequency is $\omega_c = W$.

$$\theta_{LP}(W; W) = -\frac{\pi}{4}.$$

- For $\omega \ll W$:

$$g_{LP}(\omega; W) = 0,$$

which is horizontal asymptote $g_{LP} = 0$, i.e., $|H_{LP}| = 1$.

$$\theta_{LP}(\omega; W) \approx -\frac{\pi}{2}.$$

iii High-pass factor

$$H_{HP}(s; \omega_c) := \frac{s}{s + \omega_c}$$

$$H_{HP}(i\omega; \omega_c) = \frac{i\omega}{i\omega + \omega_c}$$

$$|H_{HP}(i\omega; \omega_c)| = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}}$$

$$g_{HP}(\omega; \omega_c) = 20 \log_{10}(\omega) - 10 \log_{10}(\omega^2 + \omega_c^2)$$

$$\theta_{HP}(\omega; \omega_c) = \frac{\pi}{2} - \arg(i\omega + \omega_c)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \ll \omega_c$:

$$g_{HP}(\omega; \omega_c) \approx 20 \log_{10}(\omega) - 20 \log_{10}(\omega_c),$$

which is slant asymptote slope 20 dB/dec, y intercept $-20 \log_{10}(\omega_c)$, and x intercept $\log_{10}(\omega_c)$.

$$\theta_{HP}(\omega; \omega_c) \approx \frac{\pi}{2}.$$

- For $\omega = \omega_c$:

$$g_{HP}(\omega_c; \omega_c) = 20 \log_{10}(\omega_c) - 10 \log_{10}(2\omega_c^2) = -10 \log_{10}(2).$$

Thus the cutoff frequency is $\omega_c = \omega_c$.

$$\theta_{HP}(\omega_c; \omega_c) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

- For $\omega \gg \omega_c$:

$$g_{HP}(\omega; \omega_c) = 0,$$

which is horizontal asymptote $g_{HP} = 0$, i.e., $|H_{HP}| = 1$.

$$\theta_{HP}(\omega; \omega_c) \approx \frac{\pi}{2} - \frac{\pi}{2} = 0.$$

iv Quadratic factor

$$H_{QD}(s; B, \omega_0) := \frac{\omega_0^2}{s^2 + Bs + \omega_0^2}$$

$$H_{QD}(i\omega; B, \omega_0) = \frac{\omega_0^2}{-\omega^2 + iB\omega + \omega_0^2}$$

$$|H_{QD}(i\omega; B, \omega_0)| = \frac{\omega_0^2}{\sqrt{B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\theta_{QD}(\omega; B, \omega_0) = -\arg(-\omega^2 + iB\omega + \omega_0^2)$$

Asymptotic Bode plot and center frequencies:

- For $\omega \ll \omega_0, B$:

$$|H_{QD}(i\omega; B, \omega_0)| \approx 1$$

$$g_{QD}(\omega; B, \omega_0) \approx 0$$

which is horizontal asymptote $g_{QD} = 0$, i.e., $|H_{QD}| = 1$.

$$\theta_{QD}(\omega; B, \omega_0) \approx \frac{\pi}{2}$$

- For $\omega = \omega_0$:

$$|H_{QD}(i\omega; B, \omega_0)| = \frac{\omega_0}{B}$$

$$g_{QD}(\omega; B, \omega_0) = 20 \log_{10}(\omega_0) - 20 \log_{10}(B)$$

which has horizontal tangent $g_{QD} = 20 \log_{10}(\omega_0) - 20 \log_{10}(B)$, i.e., $|H_{QD}| = \frac{\omega_0}{B}$.

- $\omega_0 > B$ overdamped response: $g_{QD} > 0$
- $\omega_0 = B$ critically damped response: $g_{QD} = 0$
- $\omega_0 < B$ underdamped response: $g_{QD} < 0$

$$\theta_{QD}(\omega; B, \omega_0) = -\frac{\pi}{2}$$

- For $\omega \gg \omega_0, B$:

$$|H_{QD}(i\omega; B, \omega_0)| = \frac{\omega_0^2}{\omega^2}$$

$$g_{QD}(\omega; B, \omega_0) = 40 \log_{10}(\omega_0) - 40 \log_{10}(\omega)$$

which is slant asymptote slope -40 dB/dec, y intercept $40 \log_{10}(\omega_0)$, and x intercept $\log_{10}(\omega_0)$.

$$\theta_{QD}(\omega; B, \omega_0) = -\pi$$

v Band-pass factor

$$H_{BP}(s; B, \omega_0) := \frac{Bs}{s^2 + Bs + \omega_0^2}$$

$$H_{BP}(i\omega; B, \omega_0) = \frac{iB\omega}{-\omega^2 + iB\omega + \omega_0^2}$$

$$|H_{BP}(i\omega; B, \omega_0)| = \frac{B\omega}{\sqrt{B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\theta_{BP}(\omega; B, \omega_0) = \frac{\pi}{2} - \arg(-\omega^2 + iB\omega + \omega_0^2)$$

Asymptotic Bode plot and center frequencies:

- For $\omega \ll \omega_0, B$:

$$|H_{BP}(i\omega; B, \omega_0)| \approx \frac{B\omega}{\omega_0^2}$$

$$g_{BP}(\omega; B, \omega_0) \approx 20 \log_{10}(\omega) + 20 \log_{10}(B) - 40 \log_{10}(\omega_0)$$

which has slope 20 dB/dec, y intercept $20 \log_{10}(B) - 40 \log_{10}(\omega_0)$, and x intercept $2 \log_{10}(\omega_0) - \log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) \approx \frac{\pi}{2}$$

- For $\omega = \omega_0$:

$$|H_{BP}(i\omega; B, \omega_0)| = 1$$

$$g_{BP}(\omega; B, \omega_0) = 0$$

which has horizontal tangent $g_{BP} = 0$, i.e., $|H_{BP}| = 1$.

$$\theta_{BP}(\omega; B, \omega_0) = 0$$

- For $\omega \gg \omega_0, B$:

$$|H_{BP}(i\omega; B, \omega_0)| \approx \frac{B}{\omega}$$

$$g_{BP}(\omega; B, \omega_0) = -20 \log_{10}(\omega) + 20 \log_{10}(B)$$

which has slope -20 dB/dec, y intercept $20 \log_{10}(B)$, and x intercept $\log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) = -\frac{\pi}{2}$$

vi Band-stop factor

$$H_{BS}(s; B, \omega_0) := \frac{s^2 + \omega_0^2}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(i\omega; B, \omega_0) = \frac{-\omega^2 + \omega_0^2}{-\omega^2 + iB\omega + \omega_0^2}$$

$$|H_{BS}(i\omega; B, \omega_0)| = \frac{-\omega^2 + \omega_0^2}{\sqrt{B^2\omega^2 + \omega^4 + \omega_0^4 - 2\omega^2\omega_0^2}}$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg\left(\frac{-\omega^2 + \omega_0^2}{-\omega^2 + iB\omega + \omega_0^2}\right)$$

Asymptotic Bode plot and center frequencies:

- For $\omega \ll \omega_0, B$:

$$|H_{BS}(i\omega; B, \omega_0)| \approx 1$$

$$g_{BS}(\omega; B, \omega_0) \approx 0$$

which is horizontal asymptote $g_{BS} = 0$, i.e., $|H_{BS}| = 1$.

$$\theta_{BS}(\omega; B, \omega_0) = 0$$

- For $\omega = \omega_0$:

$$|H_{BS}(i\omega; B, \omega_0)| = 0$$

$$g_{BS}(\omega; B, \omega_0) = -\infty$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg(0) = \text{undefined}$$

- For $\omega \gg \omega_0, B$:

$$|H_{BS}(i\omega; B, \omega_0)| \approx 1$$

$$g_{BS}(\omega; B, \omega_0) = 0$$

which is horizontal asymptote $g_{BS} = 0$, i.e., $|H_{BS}| = 1$.

$$\theta_{BS}(\omega; B, \omega_0) = 0$$